ER. SAMRIDHA SHRESTHA

Hattigauda, Kathmandu | P.N: 9818716111, 9748300307 | samridhashrestha10@gmail.com |
| github.com/Nitride27 | Portfolio: samridhashrestha.com.np |

SUMMARY

Software Engineer at Octacore Solutions and Computer Engineering graduate (2026) working across full-stack web and applied AI: e-commerce, chatbots, handwriting recognition for KYC documents and deep learning for geospatial mapping, using AI tools and automation to ship quickly.

EXPERIENCE

SOFTWARE ENGINEER June 2026 - Present

Octacore Solutions, Kathmandu (Intern from June 2026; full-time since August 2026)

- **E-commerce:** Built backend APIs and frontend features for creationtextile.com.np, a Django and Next.js store, and a RAG customer-support chatbot using hybrid BM25 + vector retrieval, a reranker and conversation history.
- **Cheers Chatbot:** Built a customer chatbot for cheers.com.np, orchestrated in n8n.
- **KYC Form Extraction:** Building the handwriting recognition module of an internal system that reads English and Nepali fields from scanned and photographed forms.
- Fine-tuning vision-language and CNN models for handwritten text and digits, and building the training data and validation checks around them (Python, PyTorch, OpenCV).

FREELANCE WEB DEVELOPER 2026

Aqua Hundred (Aqua Minerals Nepal Pvt. Ltd.), aquahundred.com.np

- Built a multi-page marketing site with scroll animations, a product carousel, multi-step order modal, Leaflet.js depot map and Messenger integration, on Cloudflare Pages.

PROJECTS

AUTOMATED LANDFILL SITE SUITABILITY MAPPING USING DEEP LEARNING 2025-2026

- Stacked 11 geospatial layers (terrain, hydrology, land use, NDVI, population, distance to roads/rivers) and trained U-Net, ResU-Net and DeepLabV3+ to classify 5 suitability levels; custom U-Net hit 96.97% accuracy, 0.92 mIoU, 0.96 Dice.
- Turned the predicted maps into ranked candidate sites with rule-based GIS filtering: school and hospital buffers, protected-area exclusion and soil constraints (PyTorch, Rasterio, QGIS).

FLOOD SUSCEPTIBILITY MAPPING OF KATHMANDU VALLEY USING DEEP LEARNING 2024-2025

- Built a composite of flood conditioning factors for Kathmandu Valley (DEM, slope, TWI, flow accumulation, CHIRPS rainfall, distance to rivers, land use) with a historical flood inventory.
- Trained a 2D CNN and a Random Forest to classify urban pluvial flood susceptibility; the CNN reached 95% accuracy with balanced class performance, outperforming the Random Forest.

ANDROID APPS ON GOOGLE PLAY 2026

- Built and published StayT (task-based app blocker) and GTA 6 Guide with React Native/Expo and TypeScript, and Anime Studio, living anime watch faces for Wear OS with a one-time in-app unlock.

EDUCATION

ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT (T.U AFFILIATED) Kalanki, Kathmandu

Bachelor of Engineering in Computer Engineering 2021-2026

ST. XAVIER'S COLLEGE Maitighar, Kathmandu

Plus Two 2019-2021

SKILLS

AI/ML: PyTorch, Scikit-Learn, CNNs, U-Net/DeepLabV3, LoRA fine-tuning, OpenCV, RAG (hybrid search, vector databases, rerankers), LLM agents, n8n

Web & Mobile: Python, TypeScript, JavaScript, HTML/CSS, Django, Next.js/React, React Native, Tailwind, GSAP, three.js

Tools: SQL, Git, QGIS, Jupyter, Leaflet, Cloudflare Pages